FACE VERIFICATION & RE-IDENTIFICATION

Technical Assessment Report

This report documents a ResNet-based face verification and gallery–probe identification pipeline, including dataset handling, preprocessing decisions, three training objectives, cosine-similarity evaluation, validation-based threshold selection, and measured results.

100
Identities

2,500
Face images

20
Held-out test identities

Key Findings

Finding	Measured result
Best verification AUC	Triplet objective: validation AUC 0.8147; test AUC 0.8454.
Lowest estimated test EER	Triplet objective: approximately 0.2310 (23.10%).
Best identification result	Triplet objective: Rank-1 47.50%; Rank-5 85.50%.

Interpretation. Among the three evaluated checkpoints, the triplet-trained model has the strongest measured test AUC, lowest estimated test EER, and highest Rank-1/Rank-5 results in these runs. These results are specific to the dataset and protocol and do not establish production-grade recognition performance.

1. Objective and Scope

The system evaluates whether two images depict the same identity (verification) and retrieves the likely identity of a probe image from a known gallery (re-identification). A ResNet18 backbone produces 512-dimensional L2-normalized embeddings. Verification compares embeddings with cosine similarity rather than comparing classifier logits.

2. Dataset and Split Protocol

Dataset: MorphFace, obtained from the public Hugging Face repository LuxPolaris/MorphFace. The selected subset contains 100 identities with 25 PNG images per identity (2,500 images). The dataset card and usage/license conditions should be reviewed before redistribution or external reuse.

Item	Protocol / value
Identity count	100
Images per identity	25
Total images	2,500
Split	Identity-disjoint train / validation / test: 60 / 20 / 20 identities
Random seed	42
Input transform	RGB; resize to 224 × 224; tensor conversion; ImageNet mean/std normalization

Split rationale. Identity-disjoint partitions test generalization to identities not seen during training. Validation pairs are used for threshold selection; test pairs are held out for final reporting.

3. Preprocessing

The preprocessing rationale is documented as five stages. The first three subsections below reproduce the agreed wording for face presence verification, cropping, and alignment.

3.1 Face presence verification

The selected dataset consists of images organized by identity. Representative images were manually inspected to verify that they contained recognizable facial content.

The inspected samples exhibit face-focused framing, with facial features occupying most of the image area.

3.2 Face cropping

Additional face-specific cropping was deliberately omitted because the inspected images already contain tightly framed faces. Applying another crop could unnecessarily truncate facial regions, including the forehead, chin, or sides of the face.

Retaining the source framing avoids introducing an additional, unvalidated transformation and preserves the facial information already present in the images. This decision is specific to the inspected dataset and does not imply that cropping is unnecessary for unconstrained images containing substantial background.

3.3 Face alignment

Explicit landmark-based face alignment was not applied. This preserves the original variation in pose, orientation, and expression rather than imposing an additional geometric transformation.

Retaining pose variation can encourage the embedding model to learn identity-discriminative representations under varied image conditions.

3.4 Resizing to model input resolution

Images are resized to 224 × 224 pixels to provide a consistent tensor shape for batching and match the ResNet18 input pipeline. Direct square resizing may alter the source aspect ratio and distort facial geometry; aspect-ratio-preserving resizing was not used in this implementation.

3.5 Normalization

Images are converted to RGB, converted to tensors, and normalized using ImageNet channel means and standard deviations. This matches the preprocessing convention used with ImageNet-pretrained ResNet weights. Normalization does not correct lighting variation, blur, occlusion, or dataset-to-deployment domain shift.

Implemented transform: RGB conversion → resize to 224×224 → tensor conversion → ImageNet normalization. Automated face detection, detector-driven cropping, and landmark alignment are not performed by the current preprocessing code.

4. Model and Training Experiments

All three experiments use the same ResNet18-based embedding model with a 512-dimensional normalized output. They differ in training objective, allowing comparison of identity classification and metric-learning approaches.

Objective	Rationale	Selected checkpoint	Best epoch
Cross-entropy	Identity classification supervision encourages features that distinguish training identities.	`resnet18_cross_entropy_best.pt`	8
Contrastive loss	Uses positive/negative pairs to pull same-identity embeddings together and push different-identity embeddings apart subject to a margin.	`resnet18_contrastive_best.pt`	11
Triplet loss	Uses anchor, positive, and negative examples to optimize relative embedding distances.	`resnet18_triplet_best.pt`	40

The submission retains the best checkpoint from each experiment. `best_model.pth` is a convenience copy of the triplet-best checkpoint for this run.

5. Verification Evaluation

Each image is passed through the embedding network. Cosine similarity between the two embeddings is the verification score. ROC/AUC is calculated from genuine and impostor scores. The operating threshold is selected on validation pairs and then applied to test pairs; the test set is not used for threshold selection.

Training objective	Validation AUC	Test AUC	Test EER (estimate)
Cross-entropy	0.8118	0.8424	0.2375 (23.75%)
Contrastive	0.7938	0.8325	0.2520 (25.20%)
Triplet	0.8147	0.8454	0.2310 (23.10%)

The run-specific threshold and additional metrics are recorded in the JSON/CSV outputs.

6. Gallery–Probe Identification

The test partition contains 20 held-out identities. Five images per identity form the gallery (100 images); the remaining 20 images per identity form 400 probes. Gallery and probe image paths do not overlap. For each probe, candidate identities are ranked using the maximum cosine similarity to that identity's gallery images.

Objective	Rank-1 accuracy	Rank-5 accuracy
Cross-entropy	0.4600 (46.00%)	0.8450 (84.50%)
Contrastive	0.3875 (38.75%)	0.8275 (82.75%)
Triplet	0.4750 (47.50%)	0.8550 (85.50%)

Triplet training gives the highest recorded Rank-1 and Rank-5 values in this comparison. Rank-1 measures whether the correct identity is ranked first; Rank-5 measures whether it appears among the top five.

7. Submission Artifacts and Reproducibility

The submission folder includes the required scripts, checkpoints, evaluation results, and this report.

Artifact	Purpose
`dataset_preparation.py`	Dataset/split preparation and audit utility.
`train.py`, `model.py`, `dataset.py`	Training entry point, embedding architecture, and data loader.
`generate_pairs.py`	Positive/negative verification pair generation.
`evaluate_pairs.py`	Computes verification scores, metrics, and plots for the retained checkpoints.
`gallery_probe.py`	Gallery–probe identification evaluation.
`roc_analysis.py`	ROC analysis utility.
`checkpoints/best_model.pth`	Required checkpoint filename; triplet-best checkpoint copy.
`checkpoints/*_best.pt`	Best checkpoint for each of the three training objectives.
`results/roc_curve.png`	ROC curve from computed similarity scores.
`results/similarity_distribution.png`	Genuine/impostor score distributions.
`results/evaluation_results.json / .csv`	Run-specific verification metrics and comparison.
`results/gallery_probe_results.csv`	Gallery–probe Rank-1/Rank-5 comparison.

7.1 Reproduction commands

Run from the project root with the virtual environment active:

```
python .\submission\dataset_preparation.py
python .\submission\generate_pairs.py
python .\submission\evaluate_pairs.py
python .\submission\gallery_probe.py
```

Evaluation uses the existing split files and retained checkpoints.

Conclusion

The experiments compare cross-entropy, contrastive, and triplet objectives under a shared ResNet18 embedding setup. In the recorded runs, the triplet checkpoint performs best on verification AUC/EER and gallery–probe Rank-1/Rank-5.